Blood Bank MERN Stack Project — Setup Commands & Code

A complete step-by-step developer guide for setting up and running a MERN-stack Blood Bank Donor registry application.

Section 1: Project Directory Structure

```
None
blood-bank/
├── backend/
│   ├── package.json
│   └── server.js (36 lines)
└── frontend/
    ├── package.json
    └── src/
        └── App.jsx (56 lines)
```

Section 2: Terminal Setup Commands

1. Terminal 1 — Backend Setup

Execute the following commands to initialize and start the Express & MongoDB backend server:

```
None
mkdir -p blood-bank/backend && cd blood-bank/backend
npm init -y
npm install express mongoose cors
node server.js
```

2. Terminal 2 — Frontend Setup

Execute the following commands in a second terminal window to set up the React frontend using Vite:h

cd blood-bank

npm create vite@latest frontend -- --template react

cd frontend

```
npm install
npm run dev
```

```
None

## Section 3: Backend Code

File: `backend/server.js` (36 lines)

```javascript
const express = require('express');
const mongoose = require('mongoose');
const cors = require('cors');

const app = express();
app.use(cors());
app.use(express.json());

mongoose.connect('mongodb://127.0.0.1:27017/bloodbank')
  .then(() => console.log('MongoDB Connected'))
  .catch(err => console.log(err));

const Donor = mongoose.model('Donor', new mongoose.Schema({
  name: String,
  bloodGroup: String,
  contact: String,
  city: String
}));

app.get('/api/donors', async (req, res) => {
  res.json(await Donor.find());
});

app.post('/api/donors', async (req, res) => {
  const donor = await Donor.create(req.body);
  res.json(donor);
});

app.delete('/api/donors/:id', async (req, res) => {
  await Donor.findByIdAndDelete(req.params.id);
  res.json({ deleted: true });
});

app.listen(5000, () => console.log('Server running on port 5000'));
```
```

Section 4: Frontend Code

File: frontend/src/App.jsx (56 lines)

None

```
import { useState, useEffect } from 'react';

export default function App() {
  const [donors, setDonors] = useState([]);
  const [form, setForm] = useState({ name: '', bloodGroup: '', contact: '',
city: '' });
  const API = 'http://localhost:5000/api/donors';

  const loadDonors = async () => {
    const res = await fetch(API);
    setDonors(await res.json());
  };

  useEffect(() => { loadDonors(); }, []);

  const addDonor = async (e) => {
    e.preventDefault();
    await fetch(API, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(form)
    });
    setForm({ name: '', bloodGroup: '', contact: '', city: '' });
    loadDonors();
  };

  const deleteDonor = async (id) => {
    await fetch(`${API}/${id}`, { method: 'DELETE' });
    loadDonors();
  };

  return (
    <div style={{ maxWidth: '500px', margin: '30px auto', fontFamily:
'sans-serif' }}>
      <h2>Blood Bank Donor Share</h2>
      <form onSubmit={addDonor} style={{ display: 'flex', flexDirection:
'column', gap: '8px' }}>
        <input placeholder="Donor Name" required value={form.name} onChange={e
=> setForm({ ...form, name: e.target.value })} />
```

```
        <input placeholder="Blood Group (e.g. O+)" required
value={form.bloodGroup} onChange={e => setForm({ ...form, bloodGroup:
e.target.value })} />
        <input placeholder="Phone / Contact" required value={form.contact}
onChange={e => setForm({ ...form, contact: e.target.value })} />
        <input placeholder="City / Hospital" required value={form.city}
onChange={e => setForm({ ...form, city: e.target.value })} />
        <button type="submit" style={{ background: '#c62828', color: '#fff',
padding: '8px', border: 'none' }}>Register Donor</button>
      </form>

      <div style={{ marginTop: '20px' }}>
        {donors.map(d => (
          <div key={d._id} style={{ border: '1px solid #ccc', padding: '10px',
marginBottom: '8px', display: 'flex', justifyContent: 'space-between' }}>
            <div>
              <strong>{d.name} ({d.bloodGroup})</strong>
              <div>{d.city} | 📞 {d.contact}</div>
            </div>
            <button onClick={() => deleteDonor(d._id)} style={{ height: '30px'
}}>Remove</button>
          </div>
        ))}
      </div>
    </div>
  );
}
```